

Appendix

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1 The solution to the differential equation (1)

Taking the following differential equation, which is referred as (1) in the main file,

$$\frac{dD(k, C)}{dC} = g_k \frac{D(k, C)}{C} \frac{V - D(k, C)}{V}, \quad (1)$$

Let D abbreviate $D(k, C)$. Then, from (1),

$$\begin{aligned} \frac{dD}{dC} &= g_k \frac{D}{C} \frac{V - D}{V} \\ \Rightarrow \frac{V}{g_k(V - D)D} dD &= \frac{1}{C} dC \\ \Rightarrow \int \frac{V}{g_k(V - D)D} dD &= \int \frac{1}{C} dC \\ \Rightarrow -\frac{\ln\left|\frac{V}{D} - 1\right|}{g_k} + c_1 &= \ln|C| + c_2 \end{aligned}$$

with c_1, c_2 integration constants. As $\left|\frac{V}{D}\right| \geq 1$ and $C > 0$, and letting $c_2 - c_1 = \ln(h_k)$,

$$\begin{aligned} \ln\left(\left(\frac{V}{D} - 1\right)^{-1/g_k}\right) &= \ln(h_k) + \ln(C) \\ \Rightarrow \left(\frac{V}{D} - 1\right)^{-1/g_k} &= h_k C \\ \Rightarrow D &= \frac{V}{1 + (h_k C)^{-g_k}}. \end{aligned}$$

Thus, the number of distinct n -grams greater or equal to k in a *corpus* of size C , for each (L, n) pair, is given by

$$D(k, C; L, n) = \frac{V(L, n)}{1 + (h_k(L, n) C)^{-g_k(L, n)}}. \quad (2)$$